Group 1 project

# DFT Study of MB dye on Graphene-Zno hybrid

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# Objective

- **Water pollution** by synthetic dyes such as Methylene Blue (MB) is a serious problem for aquatic ecosystems and human health.
- Conventional treatment methods often fail to **fully degrade MB**, so advanced photocatalysts are needed.
- **Graphene–ZnO (G-ZnO) nanohybrids** are promising because graphene improves charge separation and extends ZnO activity under light.

# Objective

- However, the photocatalytic mechanism at the atomic level, and **how structural changes affect performance**, are not yet well understood.

- This work uses high-performance computing, **quantum-chemical simulations (DFT) using quantum espresso to study and predict MB degradation on G-ZnO and to guide the design of better photocatalysts**

# Technology used

- **High performance Computing**

  Linux
  Slurm

- **Quantum Simulation Tools**

  Density Functional theory
  Quantum ESPRESSO

- **Python libraries**

  ASE
  py3Dmol
  multiprocessing

# High performance computing

- HPC uses clusters of many interconnected computers to solve problems that are too large or slow for a normal laptop/desktop.
- Enables parallel computing, where many processors work on different parts of a problem at the same time to speed up calculations.
- Widely used in research and industry, including weather forecasting, materials science/DFT, drug discovery, AI, and big-data analysis.

# Density functional theory

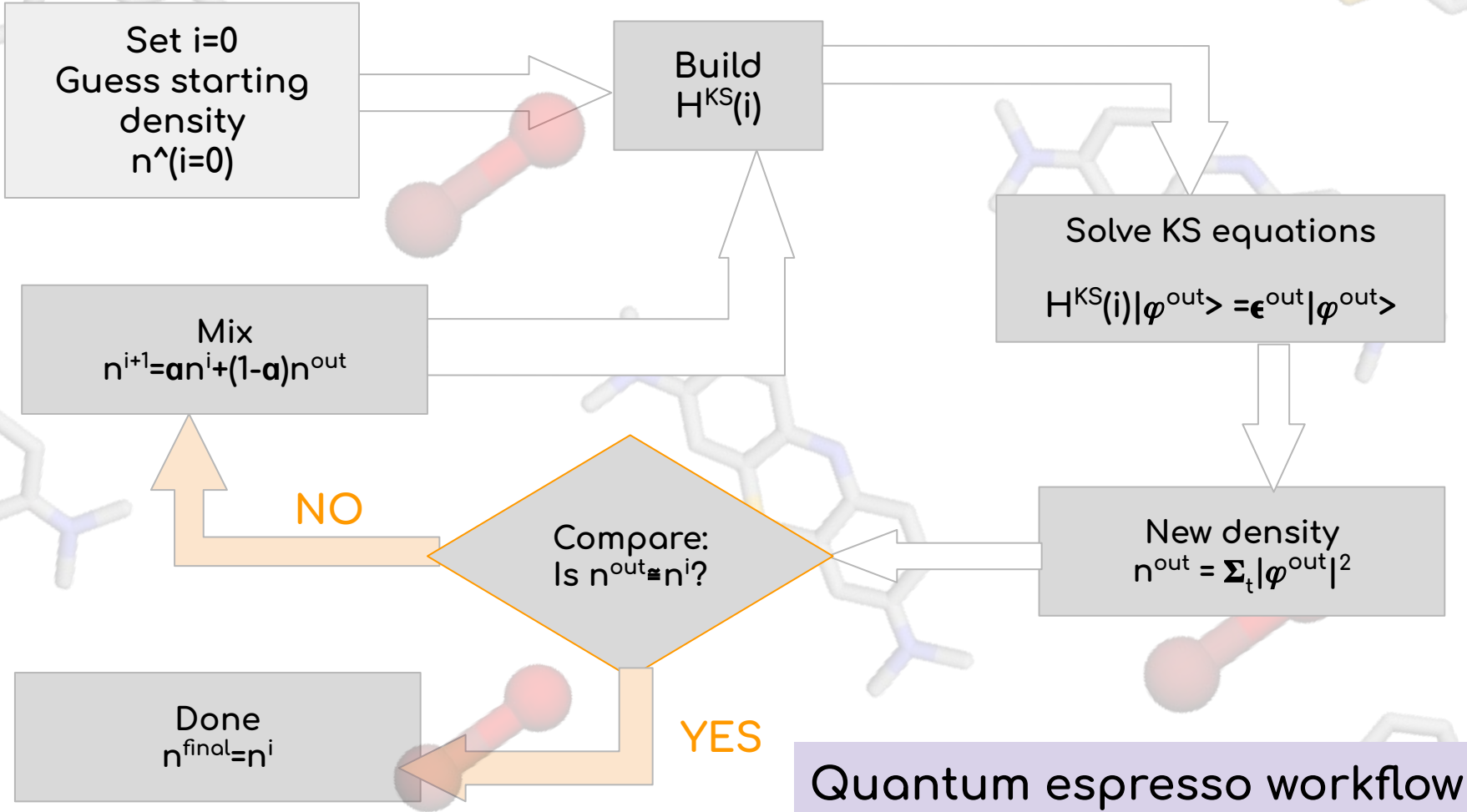
- DFT is a **quantum-mechanical method** used to study the electronic structure of atoms, molecules, and solids.

- It is based on the idea that **all properties of a many-electron system are determined by electron density**, not the many-electron wavefunction.

- The **total energy functional** includes: kinetic energy + electron–nuclear interaction + electron–electron interaction + exchange–correlation term.

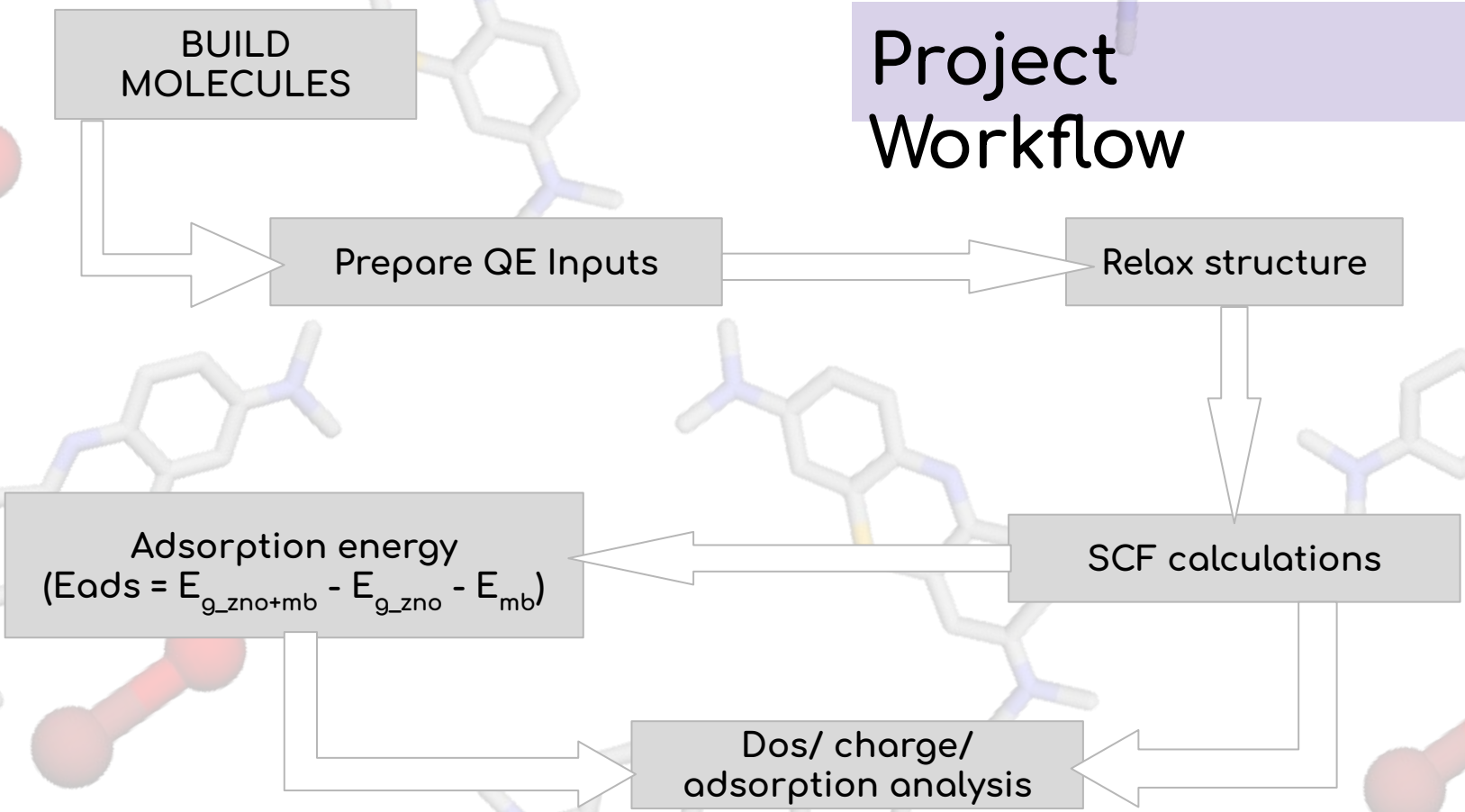
# Quantum espresso

- **Open-source suite for electronic-structure calculations** and materials modeling using density functional theory, plane waves, and pseudopotentials.
- Supports geometry optimization, band structure, phonons, molecular dynamics, **adsorption studies**, and many other condensed-matter simulations.
- Efficiently **runs on parallel architectures** using MPI/OpenMP, making it suitable for large systems and supercomputers.

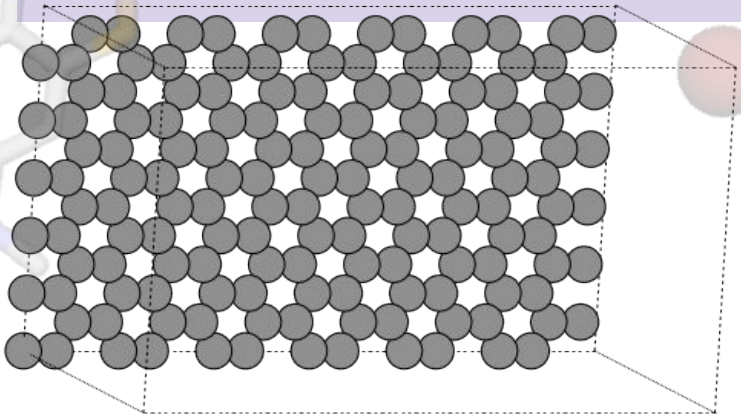



# Quantum espresso workflow

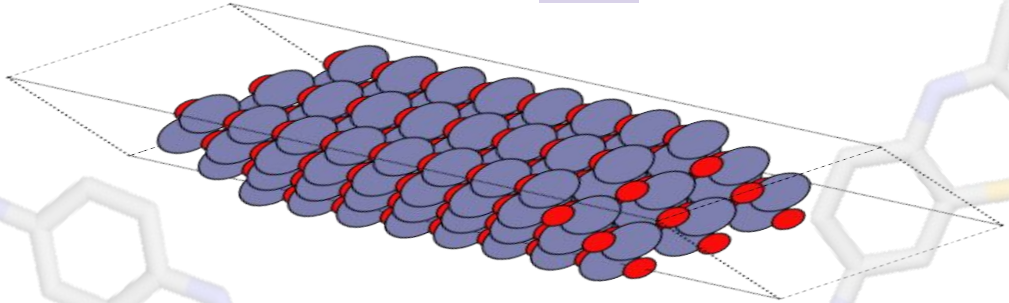
# Project Workflow



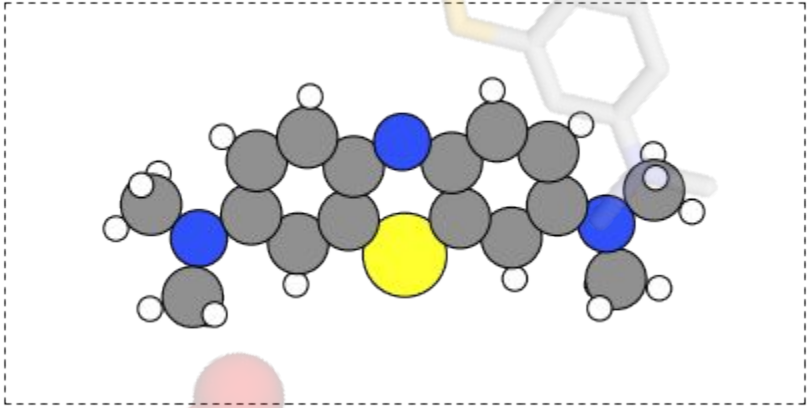
# System overview



Zno

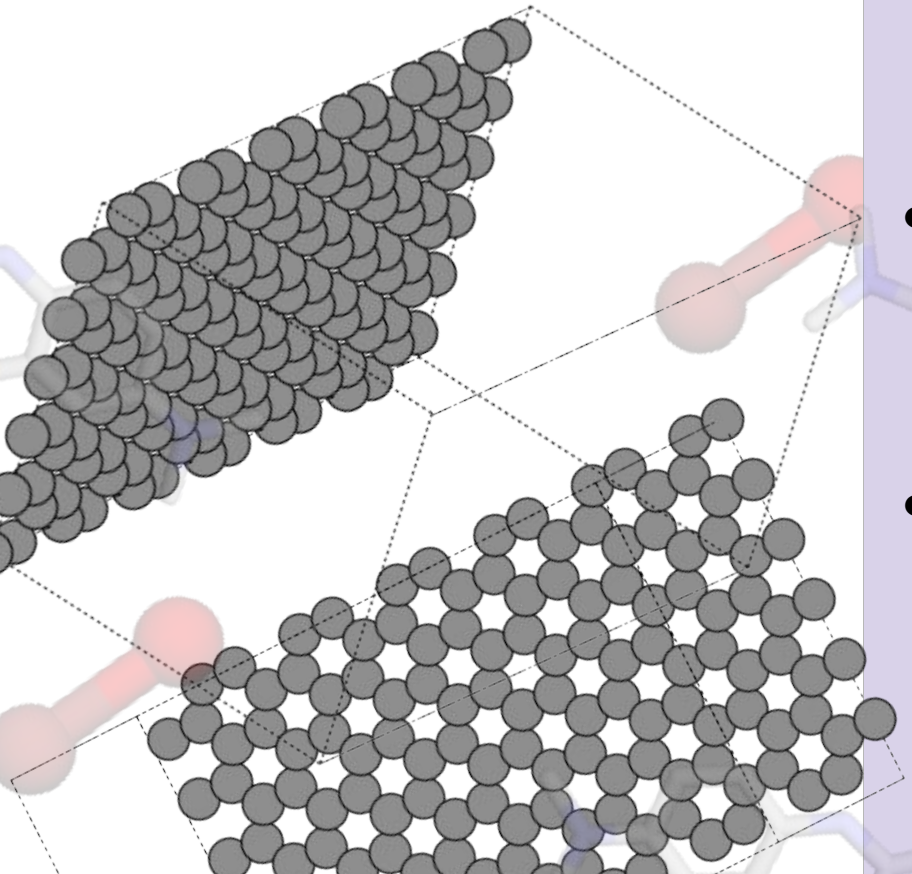


Graphene sheet



Methylene blue dye

# Graphene

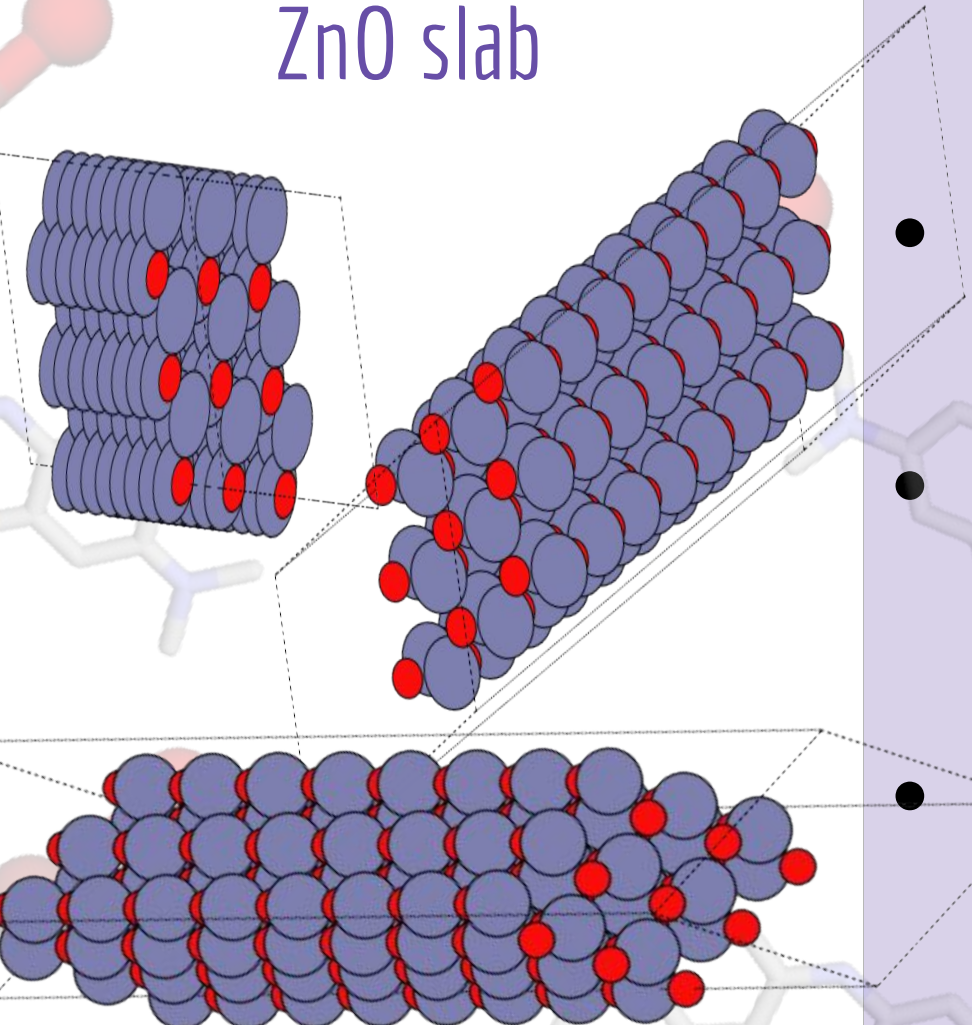


- Graphene is a single-atom-thick, two-dimensional sheet of $sp^2$-bonded carbon atoms arranged in a hexagona honeycomb lattice.

- Graphene was constructed using a rectangular 4-atom unit cell with a C–C bond length of 1.42 Å, consistent with experimental values.

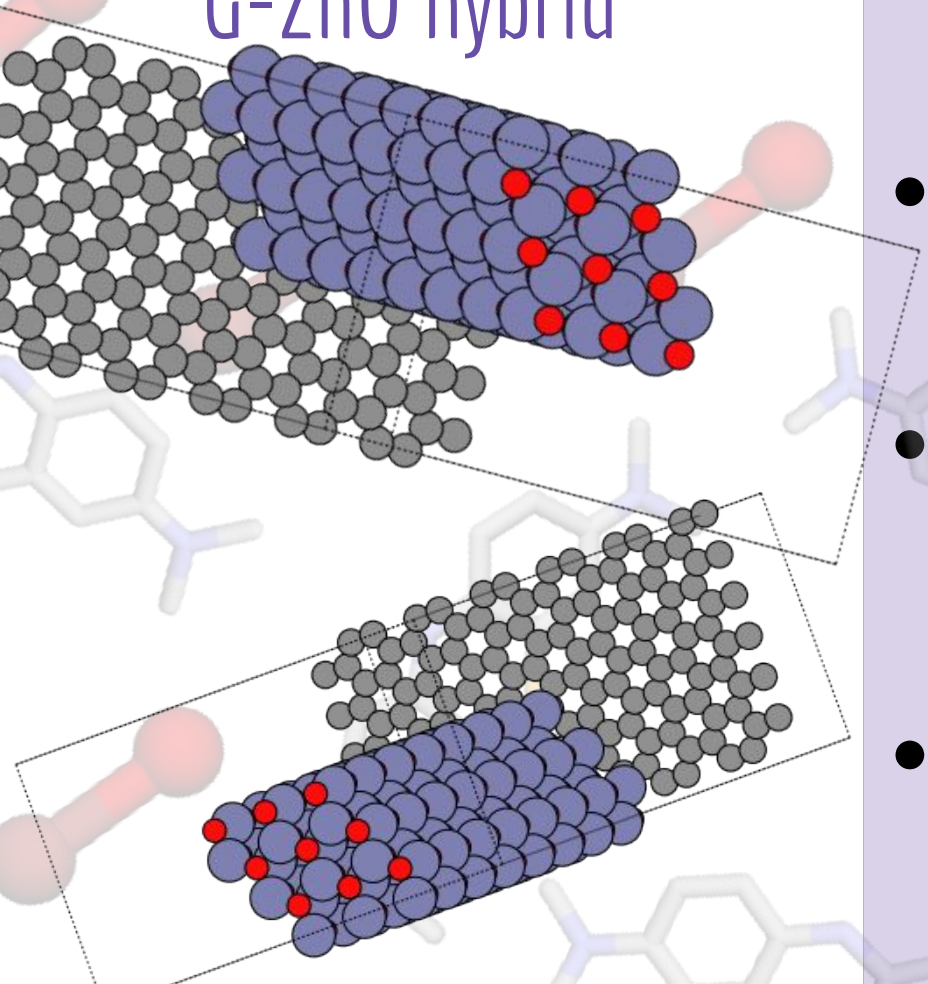
- A 6 × 6 supercell (144 carbon atoms) wa generated with periodic boundary conditions in x and y and a 15 Å vacuu along z to model an isolated graphene sheet.

# ZnO slab



- ZnO was modeled in the wurtzite cryst structure using experimental-like lattic parameters (a = 3.25 Å, c = 5.20 Å).
- Zn–O bonding in wurtzite ZnO is $sp^3$-li tetrahedral coordination (4 Zn–O bond per atom)
- A (0001) surface slab was constructed with 8 atomic layers and a 15 Å vacuum along the surface normal to avoid periodic interactions.
- The slab was centered along the z-direction and expanded to a 3 × 3 in-plane supercell, making it suitable f surface and adsorption studies.

# G-ZnO hybrid



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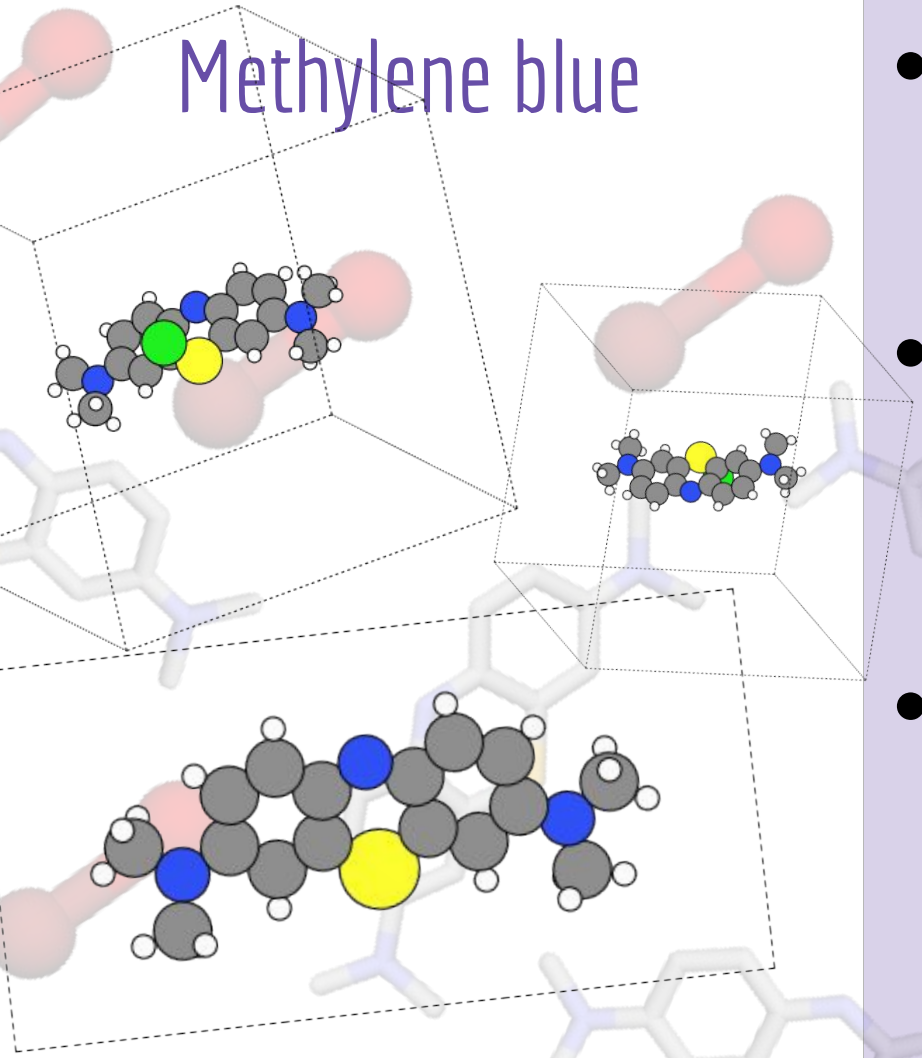
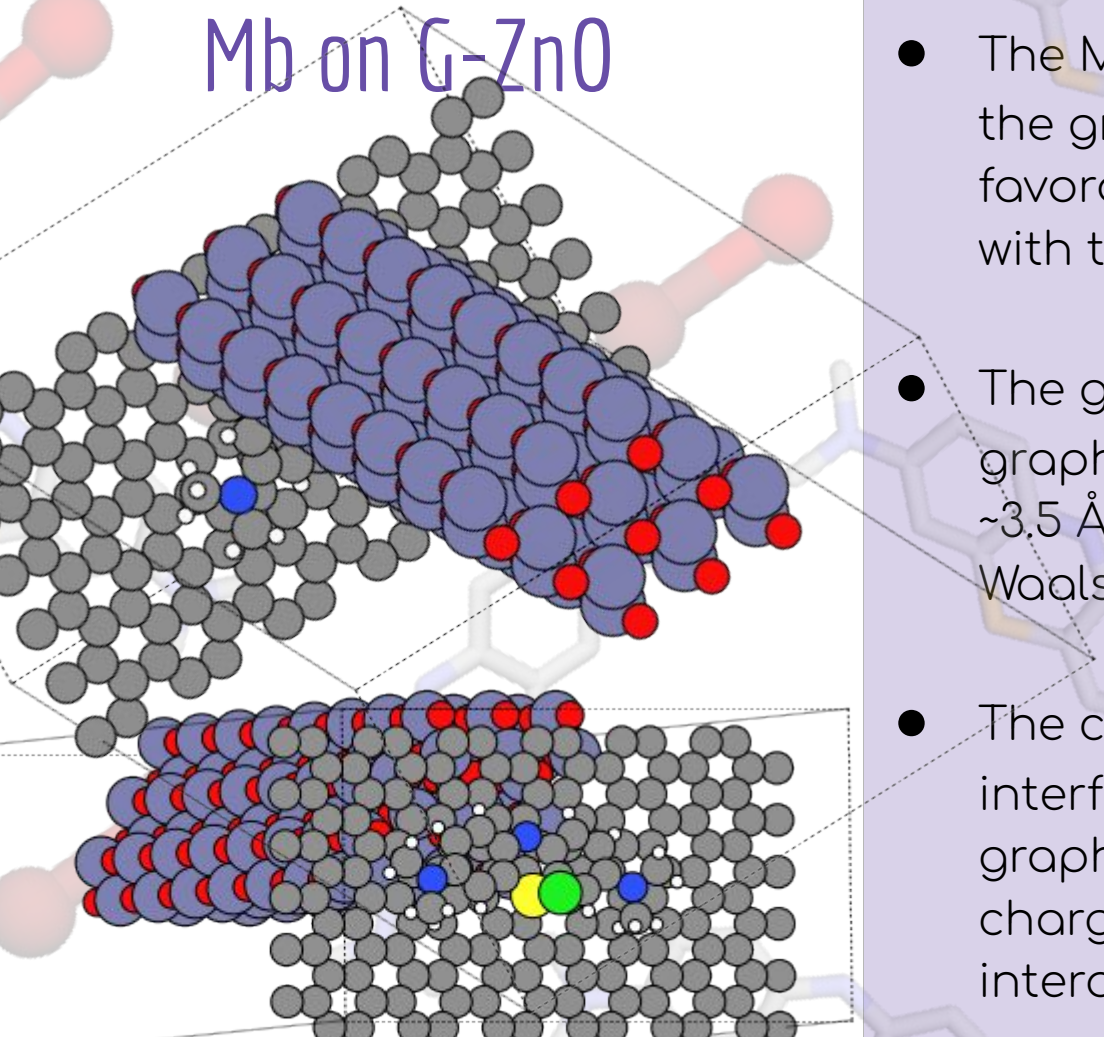
# Methylene blue



- Methylene blue is a cationic organic d with a planar aromatic structure and a +1 formal charge.
- The MB molecule was constructed fron its SMILES representation, converted into a 3D geometry, and structurally optimized.
- MB exhibits strong π-conjugation and visible-light absorption, which makes i a commonly used model dye in photocatalytic and adsorption studies

# Mb on G-ZnO



- The MB molecule is positioned parallel to the graphene surface, indicating a favorable orientation for π–π interaction with the graphene sheet.
- The g-ZnO component is placed above graphene with a physical separation of ~3.5 Å, ensuring non-covalent (van der Waals) interaction at the initial stage.
- The combined structure forms a hybrid interface where MB is exposed to both graphene and ZnO, enabling potential charge transfer and photocatalytic interaction.

# RESULTS

```
negative rho (up, down):  8.970E-03 0.000E+00

total cpu time spent up to now is     5900.2 secs

total energy              =    -546.34798365 Ry
estimated scf accuracy    <       0.01886005 Ry

End of self-consistent calculation

convergence NOT achieved after  60 iterations: stopping
```

Mb scf energy did not converge

# RESULTS

```
     the Fermi energy is    -1.6856 ev

!    total energy              =   -2655.86997810 Ry

     total all-electron energy =    -10973.899977 Ry
     estimated scf accuracy    <          2.3E-09 Ry
     smearing contrib. (-TS)   =       0.00032762 Ry
     internal energy E=F+TS    =   -2655.87030571 Ry

     The total energy is F=E-TS. E is the sum of the following terms:
     one-electron contribution =  -11349.45973993 Ry
     hartree contribution      =    5791.90886665 Ry
     xc contribution           =    -668.55341067 Ry
     ewald contribution        =    4458.29824371 Ry
     one-center paw contrib.   =    -888.06426547 Ry

     convergence has been achieved in  12 iterations
```

Graphene relax converges after 12 iteration

# RESULTS

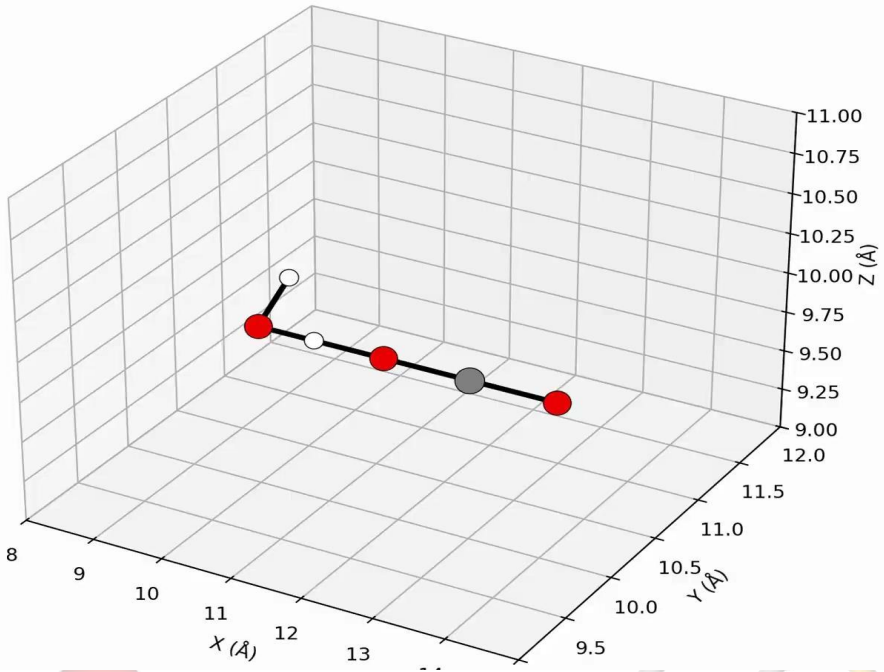
```
the Fermi energy is     7.9139 ev

WARNING: integrated charge=   972.00315922, expected=   972.00000000

!    total energy              =  -15414.03610426 Ry

     total all-electron energy =   -202105.207407 Ry
     estimated scf accuracy    <       0.00000164 Ry
     smearing contrib. (-TS)   =       0.00020074 Ry
     internal energy E=F+TS    =  -15414.03630500 Ry

     The total energy is F=E-TS. E is the sum of the following terms:
     one-electron contribution =   -4066.89568293 Ry
     hartree contribution      =    3161.80267088 Ry
     xc contribution           =   -2958.71579655 Ry
     ewald contribution        =   -5880.45468466 Ry
     one-center paw contrib.   =   -5669.77281175 Ry

     convergence has been achieved in  11 iterations
```

ZnO scf converge after 11 iteration

# RESULTS

| | Self -consistent energy | Relax energy |
|---|---|---|
| Graphene | | -2655.86997810 Ry |
| ZnO | -15414.03610426 Ry | |
| G-ZnO | | |
| Mb Dye | | |
| Mb dye on G-ZnO | | |

# Expected results



# Conclusion

Although numerical convergence was not achieved within the time and resource constraints, the study provided important insight into **model preparation**, parameter sensitivity, and the **challenges associated with DFT** simulation of dye–semiconductor hybrids. Future work will include **dispersion-corrected functionals**, larger cutoffs, and hybrid functional validation.

# Thank You